

Tutorial 10 Questions

Finite State Machine Design 2

Please try out these self-study questions (labeled “SS”). These will not be discussed in class and solutions will be provided later.

SS1. Implement the circuit of Tutorial 9, Q3 using D flip-flops and minimum logic gates.

SS2. Implement the circuit of Tutorial 9, Q4 using D flip-flops and minimum logic gates.

Q1. A digital circuit is to be designed to implement the “snooze” function of an alarm clock. The circuit’s input and output signals are as follows:

Inputs: RING, RESET, CLK

Output: ALARM

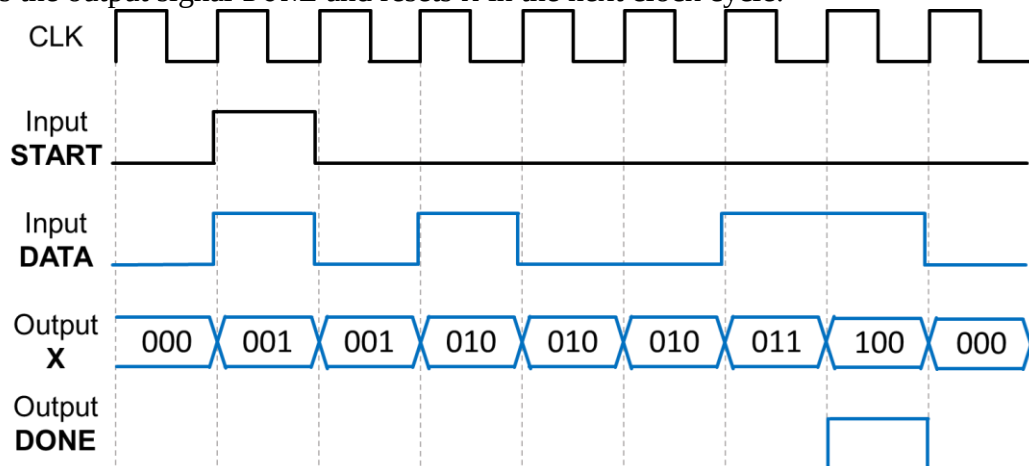
The circuit remains in the “wait” mode until input RING is detected to be **TRUE**. When this happens, the circuit goes into the “snooze” mode by ringing the ALARM for 10 seconds, waiting for 1 minute and continuously repeats this cycle until the user resets the circuit causing RESET to be **TRUE** for one clock period.

Derive the state transition diagram, assuming that an accurate 0.1Hz clock is available as an input. To simplify the state transition diagram, adopt a suitable architectural element as a submachine. Write a Verilog program which realizes this state machine.

Q2. A digital circuit is designed to find the number of 1’s in a serial input DATA signal.

Once the input START becomes **TRUE**, 7 serial bits are provided consecutively into the circuit via the 1-bit input signal DATA, one-bit per clock cycle.

The circuit evaluates DATA and continuously updates X, a 3-bit output signal. X reflects the total number of 1’s received up to the current clock cycle. After the 7-bits have been received, the circuit asserts the output signal DONE and resets X in the next clock cycle.



Derive the state transition diagram for this circuit. Write a Verilog program which realizes the circuit and verify its performance using available simulation tools.

- Q3. A synchronous state machine needs to be designed to *count people entering and leaving a room*. The room has a *separate* entrance and exit.

Each door is equipped with a sensor to detect people *coming in through the entrance* or *leaving through the exit*. The sensors provide signals ENTER (when a person enters) and LEAVE (when a person exits), which are **TRUE** for one clock period.

The entrance and exit are each narrow enough for only one person to pass through at a time, but it is possible for one person to enter *while* another is exiting during one clock period.

The machine should output an **EMPTY** signal when nobody is in the room and a **FULL** signal when there are 3 people (*the maximum allowed*) in the room. Nobody will be allowed into the room while **FULL** is asserted.

The **EMPTY and **FULL** signals should be asserted as soon as the conditions are satisfied, i.e. before the next active clock transition.**

Derive the state transition diagram and write a Verilog program which realizes this state machine. You may assume that an accurate 0.5Hz clock is available as input.